

Question bank

Electronic Principles and Circuits (BEC303)

Module 1

- 1) Explain Voltage divider bias using (i) Approximate analysis (ii) Exact analysis
- 2) Write explanatory notes on (i) Coupling Capacitor (ii) Bypass Capacitor
- 3) With a neat circuit diagram and necessary equations explain the operation of Base Biased Amplifier
- 4) What is a stiff voltage divider? With a neat circuit diagram and necessary equations explain the operation of Emitter Biased Amplifier (Voltage Divider Bias Amplifier)
- 5) Explain the following terms related to amplifier (i) Voltage Gain (ii) 10 percent rule for small signal operation (iii) Current gain and (iv) ac emitter resistance
- 6) Explain transistor models used for ac analysis.
- 7) Explain the steps involved in the analysis of CE amplifiers.
- 8) Draw the ac equivalent circuit for a base biased amplifier and determine the voltage gain using (i) T - Model (ii) π - Model
- 9) Draw the ac equivalent circuit for a Emitter bias or VDB amplifier and determine the voltage gain using (i) T - Model (ii) π - Model
- 10) Explain the Loading Effect of Input Impedance on transistor CE amplifier.
- 11) With a neat circuit diagram and equations, explain Common Collector (CC) Amplifier and determine its (i) Voltage gain (ii) AC resistance (iii) input impedance (iv) output impedance.

Module 2

1. Explain the voltage divider bias scheme for biasing MOSFET.
2. State the disadvantage of fixed VGS biasing technique and explain how stability of operating point is achieved in drain to gate feedback resistor biasing technique in a MOSFET amplifier.

3. Explain the MOSFET biasing technique using a large drain-to-gate feedback resistance R_G .
4. Draw the conceptual MOSFET amplifier circuit and derive the expression for (i) small signal drain current i_D (ii) Transconductance g_m and (iii) voltage gain A_v
5. With the small signal equivalent model of MOSFET, derive an expression for voltage gain and transconductance.
6. Draw and explain the small signal model of the MOSFET amplifier, assuming $\lambda \neq 0$.
7. Mention the design parameters for a MOSFET amplifier and how transconductance can be computed using these parameters.
8. What is transconductance of MOSFET. Mention the three different expressions used to calculate the transconductance.
9. Explain the various MOSFET amplifier configurations and derive the expression for characterizing parameters of MOSFET amplifier
10. Define the terms input resistance, output resistance, voltage gain and overall gain of an amplifier.
11. Explain the common source amplifier without source resistance and derive the necessary equations.
12. With a neat circuit diagram and ac equivalent circuit derive the expressions for R_{in} , A_{vo} , A_v and R_o for common source amplifier with an un-bypassed source resistance.
13. Explain the working of common source amplifier with source resistance and derive its characterizing parameters.
14. With a neat circuit diagram and ac equivalent circuit, derive the expression for R_{in} , A_v , and R_o with bypassed source resistance.
15. Obtain the expression for R_{in} , A_v , and R_o for a Common Drain amplifier or source follower.
16. With proper equivalent circuits, explain the common drain MOSFET amplifier and derive its characterizing parameters.

17. Analyze the circuit of common drain amplifier and derive the expressions for no-load voltage gain, overall voltage gain, input resistance and output resistance.
18. With a neat circuit diagram and ac equivalent circuit, derive the expressions for R_{in} , A_{vo} , A_v and R_o for a Source follower.
19. With a neat circuit diagram and ac equivalent circuit, explain Common gate amplifier.

Module 3

1. Explain the working of three input inverting summer and show how it can be used as an averaging circuit.
2. Explain the working of two input op-amp Subtractor circuit
3. Explain the operation of Summing amplifier which has both inverting and non-inverting inputs with equations and show that $V_O = V_{O1} + V_{O2} = V_3 + V_4 - V_1 - V_2$
4. With a neat circuit diagrams and equations, explain the working of Binary weighted DAC .
5. With a neat circuit diagrams and equations, explain the working of R-2R ladder DAC.
6. What is comparator? Explain the different types of comparators without and with reference.
7. Explain the operation of inverting and non inverting comparators with reference.
8. With a neat circuit diagram and necessary equations, explain the operation of Schmitt trigger circuit.
9. What is a regenerative comparator? With a neat circuit diagram and equations, explain inverting Schmitt trigger.
10. Draw and explain the functional diagram of 555 timer.
11. Explain with a neat diagrams and waveforms the 555 timer as Monostable multivibrator.
12. Draw the internal schematic of IC 555 timer and configure it for Monostable operation and explain its working with necessary equations.
13. Explain the working of a Monostable multivibrator using 555 timer with a neat functional diagram and wave forms. Derive the equation for its pulse width.

14. Explain with neat diagrams and waveforms the 555 timer as Astable multivibrator.
15. Draw the internal diagram of IC 555, configure it for Astable operation and explain with necessary equations and waveforms.
16. What is an Oscillator? Explain the principle of oscillator.
17. State the conditions for generating sustained Oscillations.
18. With neat circuit diagram, explain the working of (i) Wien bridge oscillator (ii) RC phase shift oscillator (iii) Colpitt's Oscillator (iv) Hartley Oscillator (v) Crystal Oscillator Write the expression for frequency.

Module 4

1. Explain the general feedback structure with relevant equations.
2. Mention the important features of negative feedback.
3. Explain the important properties of negative feedback used in amplifiers.
4. Explain the classification of feedback amplifiers or explain the various feedback topologies.
5. With a neat circuit diagram explain (i) VCVS amplifier (ii) VCIS amplifier (iii) ICIS amplifier and (iv) ICVS amplifier
6. With a neat diagram, derive the expression for (i) Closed loop gain (ii) input resistance with feedback (iii) output resistance with feedback for VCVS amplifier.
7. What are active filters? Mention the advantages of active filters over passive filters ?
8. Draw the ideal responses. of four basic filters and Identify the passband, stop-band, and cutoff frequencies of each.
9. What is an all-pass filter? What is its significance?
10. Explain the operation of first order (i) low pass filter (ii) Highpass filter with neat circuit diagram and frequency response.
11. Explain the operation of (i) VCVS Unity-Gain Second- Order Low-Pass Filter (ii) VCVS Equal Component Low Pass Filter.
12. Explain the operation of (i) VCVS Unity-Gain Second- Order High-Pass Filter (ii) VCVS Equal Component High Pass Filter

13. Show a bandpass filter can be constructed by the use of low pass filter and a high pass filter.
14. Sketch the expected frequency response and explain the operation of a single stage bandpass filter
15. Explain the operation of (i) MFB Bandpass Pass Filter (ii) Bandstop filter (notch filter).

Module 5

1. What is an amplifier? Explain the classification of amplifiers based on Class of Operation.
2. What is an amplifier? Explain the classification of power amplifiers.
3. Write short notes on Types of coupling used in amplifiers.
4. What is a load line? Draw and explain the concept of DC load line and AC load line for a transistor using voltage divider bias showing the end points.
5. What is a Class A power amplifier? Explain the operation of class A amplifier and derive the expression for its power gain, power dissipation, current drain and its efficiency.
6. With a neat circuit diagram, explain the operation of (i) Class B push-pull amplifier (ii) Class B push-pull emitter follower.
7. What is a Class C Power amplifier? With a neat circuit diagram explain its operation and also the DC clamping in Class C operation.
8. What is a Thyristor? Mention few applications of Thyristors.
9. With proper illustrations, explain the operation of Four layered diode.
10. Explain the terms Break-over and Low Current drop out with respect to Four layered diode.
11. With a neat diagram explain the operation Silicon Controlled Rectifier (SCR).
12. What is Phase control? With a neat circuit diagram and equations, explain SCR phase control circuit
13. Define the terms (i) Holding current and (ii) Latching current
14. Write short notes on (i) Resetting SCR (Commutation) (ii) Diac (iii) Triac
15. With a neat diagram, explain the operation of IGBT

16. Write explanatory note on (i) Photo-SCR (ii) Gate Controlled Switch (iii) Silicon Controlled Switch
17. With a neat diagram, explain the working of UJT relaxation oscillator